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RESEARCH PAPER

A Theoretical Framework and The Development of a Multidimensional Scale of Purpose in Life

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Abstract
Purpose in life is a key resource in existential psychology. However, its conceptual variability has hindered its application in mainstream practice. This paper presents a holistic, theoretical framework of purpose in life, the purpose in life system, developed from an integration of existing motivational theories and models. Based on our framework, we constructed the 12-item Multidimensional Measure of Purpose in Life (MMOP) and tested its psychometric properties across two studies. In Study 1 (N = 238), confirmatory factor analysis supported a three-factor higher-order model of purpose in life (present-oriented purpose in life , future-oriented purpose in life, purpose congruence across time), with all fit indices meeting recommended thresholds and all subscales demonstrating high internal consistency reliability. Study 2 (N = 286) examined associations between the MMOP and established measures of purpose in life, motivation, and mental health, providing evidence for the MMOP's concurrent, convergent, and discriminant validity. While further refinement of the MMOP is recommended in future research, the findings offer promising preliminary support for our integrative, multidimensional measure of purpose in life.

Keywords Measure · Purpose in Life · Psychometric analysis · Scale Development · Multidimensional Assessment · Motivation

1 Introduction

Existential psychology examines how individuals experience and navigate fundamental aspects of existence (Vos, 2023). While themes such as mortality, responsibility, and suffering are central to this field (Frankl, 1984; Yalom, 1980), purpose in life is arguably of utmost

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importance to existential scholars and clinicians. Rooted in ancient philosophy, purpose in life is widely regarded as essential to psychological well-being, with its contemporary therapeutic application dating back nearly a century (Park & Peterson, 2009; Wadlington, 2012). Frankl (1984), a pioneering existentialist, emphasised the importance of uncovering purpose in life as a source of resilience and a means of overcoming life's adversities, most notably through logotherapy, Frankl's therapeutic approach centered on the quest for meaning. Similarly, Yalom (1980) identified purpose in life as a central resource in existential therapy through which people can explore and understand other existential concerns.

Despite its significance, purpose in life theories are highly variable. For instance, some authors have described it as a sense of direction (Ryff & Keys, 1995), while others have viewed purpose in life as the ability to recognise current behaviours as contributing to future outcomes (Stillman & Baumeister, 2009), as the identification of meaningful goals (Kosine et al., 2008), or as entwined with identity development and self-transcendence (Bronk, 2011; Damon et al., 2003). This conceptual variability has contributed to a lack of structure and unclear parameters in existential therapy, hindering its empirical support and utilisation in mainstream practice (Correia et al., 2017; Keshen, 2006).

We argue that no single perspective fully accounts for the vast influence of purpose in life across time and contexts. However, integrating these perspectives will likely offer a more comprehensive understanding of this construct. To advance existential psychology and therapy, this paper introduces the purpose in life system, a theoretical framework of purpose in life developed through a synthesis of existing motivational perspectives. We then present a new measure of purpose in life derived from this new framework to support future research in this area.

2 Conceptualising Purpose in Life

Our theoretical framework is that purpose in life is a dynamic internal motivational system in which people's sense of coherence and the significance of their unique history inform their life orientations to promote meaningfulness in the present and into the future. An illustrative model of the purpose in life system is presented in Fig. 1 below.

As seen in Fig. 1, we suggest that purpose in life occurs as a process involving past, present, and future-oriented components. Past oriented processes contribute to a sense of significance and coherence of met and unmet needs by organising autobiographical memories into a meaningful personal history. These historical interpretations shape present orientations influencing what people find rewarding and worthy of engagement. Present orientations, in turn, informs future-oriented processes and goal-motivational processes. Future oriented processes then guide meaning-making and behaviour in the present while also retrospectively shaping interpretations of the past. The purpose in life system therefore draws from a holistic perspective inspired by Gestalt principles (see Wertheimer, 2010), in which purpose in life emerges from a dynamic integration of past, present, and future orientations and cannot be explained by any single component. Each stage is informed by well-established motivational theories, which are elaborated in the following sections.

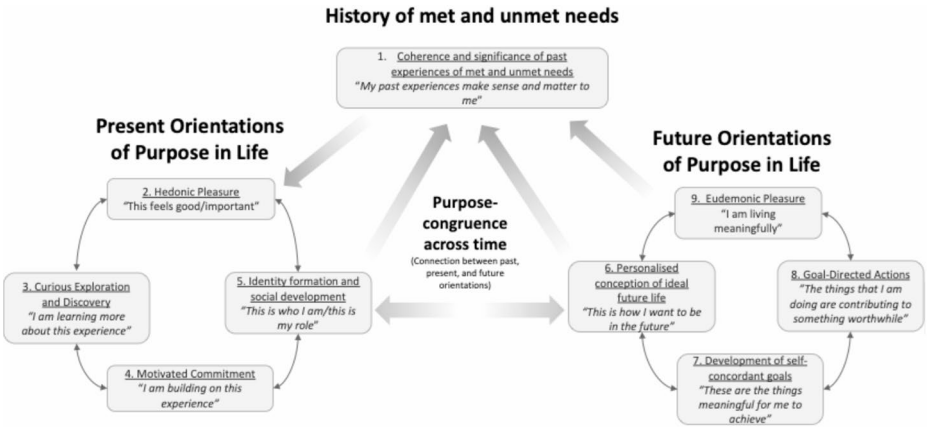


Fig. 1 The Purpose in life System

2.1 Purpose in Life in the Past

Our theoretical framework highlights how past experiences of met and unmet psychological needs shape the development of purpose in life. This aspect of our framework is grounded in Deci and Ryan's (2000) self-determination theory as well as approach-avoidance motivational principles (Elliot & Thrash, 2002). In line with self-determination theory, we posit that psychological needs encompass autonomy (experience of acting with volition), competence (the experience of effectiveness and mastery), and relatedness (the experience of connection and belonging) are fundamental to psychological growth and development (Deci & Ryan, 2000; Vansteenkiste & Ryan, 2013). These needs form the foundation of psychological growth, and their satisfaction or frustration generate emotionally salient experiences that guide attention, interpretation, and behaviour (Deci & Ryan, 2000; Dizén et al., 2005).

Positive emotional responses reinforce behaviours that prolong or repeat events that support psychological needs, whereas negative emotional experiences prompt behaviours aimed at avoiding situations that thwart these psychological needs (Deci & Ryan, 2000; Dizén et al., 2005). These approach and avoidance responses illustrate how psychological needs motivate and influence peoples' life direction (Elliot & Thrash, 2002; Vansteenkiste & Ryan, 2013).

However, individuals differ in how they experience the satisfaction and frustration of psychological needs, as these experiences are shaped by their unique life histories (Vansteenkiste & Ryan, 2013). Therefore, purpose in life development requires more than needs satisfaction alone; it also requires a coherent understanding of how these needs were met and unmet. This understanding helps people to identify their personal direction for growth and attribute significance to past experiences which in turn informs long-term life directions (Deci & Ryan, 2000; Vansteenkiste & Ryan, 2013).

In sum, psychological needs are essential for human development (Deci & Ryan, 2000). Within people's unique history of met and unmet psychological needs, is contextual information on how their psychological needs have been satisfied or thwarted in the past (Vansteenkiste & Ryan, 2013). Having a coherent understanding of these past need experiences provides insight into the conditions that foster personal growth and help avoid future

need frustrations (Vansteenkiste & Ryan, 2013). Recognising the significance of these experiences motivates people to sustain favourable conditions over time (Deci & Ryan, 2000; Vansteenkiste & Ryan, 2013), which is necessary for the promotion of purpose in life. Our framework positions coherence and significance of a person's history of psychological needs as a foundational element of their purpose in life.

For instance, a person who has consistently experienced autonomy support, may develop purpose in life centred on self-directed growth given their history conveys that pursuing goals grounded personal values is both possible and rewarding. In contrast, an individual whose autonomy was repeatedly constrained may frame their purpose in life based on resisting perceived constraints on themself and others. In both cases, the history of autonomy satisfaction or frustration informs a way of navigating life that is purposeful, because it protects and sustains the sense of autonomy ongoing.

In some more specific examples, those who have histories of feeling capable and competent in academic contexts, may be more inclined to form purpose that involves intellectual engagement. Whereas those who have always struggled in academia may direct their sense of competence to other areas such as more hands-on work, sports, social networking, or other areas of passion. Likewise, an individual who has enjoyed warm cohesive connection with their family based on traditional practices may later find purpose in maintaining these traditions with their own family. Those who experience these practices as rigid or alienating may avoid them, seeking alternative ways of fostering connection.

2.2 Purpose in Life in the Present

The present-oriented processes in our framework were derived by synthesising constructs that consistently appear in theoretical and empirical literature on purpose, motivation, and identity development. Hedonic pleasure, curious exploration, and motivated commitment repeatedly emerged as mechanisms through which people engage with and deepen their involvement with a meaningful life direction. These processes promote meaningful identity formation and social development, and are informed by well-established theories, including work on hedonic pleasure (Alba & Williams, 2013; Huta, 2015), Fredrickson's (2004) broaden and build theory, Kashdan et al.'s (2004) research on curiosity and exploration, and theories linking purpose in life and identity (Bronk, 2011; Hill et al., 2023). To meaningfully organise these constructs in a theoretically informed manner, we referred to Schnell's (2009) hierarchical model of meaning, which posits a bottom-up pathway involved in purpose in life from perceptions (hedonic pleasure), actions (curious exploration), goals (motivated commitment), and sources of meaning (identity and social role development).

Integrating these perspectives, our the purpose in life system suggests that people who engage effectively in present-orientated purpose are attuned to feelings of *hedonic pleasure*, the positive, transient, and present-orientated emotional states such as happiness, fulfilment, comfort, and satisfaction (Huta, 2015). Hedonic pleasure serves as a signal to promote *curious exploration and discovery*, the process of seeking, experimenting, and uncovering novel experiences with an inquisitive and open mindset (Alba & Williams, 2013; Fredrickson, 2004; Huta, 2015; Kashdan et al., 2004). Through curious exploration, people broaden their perspective, allowing them to discover aspects of life that resonate with what is personally meaningful (Kashdan et al., 2004). When something meaningful is discovered, this can become a target for *motivated commitment*, defined here as the sustained investment

of resources (e.g., energy, effort, time) into a meaningful aspect of life. Repeated and prolonged engagement with these meaningful life domains enables people to integrate them into their sense of self, contributing to *identity formation and social role development*, the understanding of self and one's role among others (Bronk, 2011; Hill et al., 2023). Greater coherence in the perceptions of self and others reinforces hedonic pleasure, reinforcing present oriented purpose in life (Thoits, 2012).

Those who struggle with present-orientated purpose in life processes may therefore feel less engaged in their daily life. This can limit opportunities for the development of a meaningful identity leaving individuals to feel disconnected, stagnant, and uncertain in their role among others.

2.3 Purpose in Life in the Future

Similar to present-oriented components of purpose in life, future-orientations were selected based on converging evidence across theories of future-directed motivation. Constructs such as current-ideal self-representations, self-concordant goals, goal-directed actions, and eudaimonic pleasure, have been repeatedly identified as mechanisms through which people pursue meaningful futures. These processes are informed by an extensive body of literature linking goal engagement to personal growth. Key theories underpinning future-oriented purpose in life include Higgin's (1987) self-discrepancy theory, self-concordance goal theory (Sheldon & Elliot, 1999), Emmons' (2003) theory of meaningful goal pursuit, Carver & Scheier's (1998) control theory, and research on eudaimonic well-being (Huta, 2015). As Emmons (2003) stated, "goals are the concretised expression of future orientation and life purpose" (p.106). In line with Schnell's (2009) hierarchical model of meaning, future-oriented constructs were organised following top-down pathway from sources of meaning (identity and social role development), goals (self-concordant goals), behaviours (goal-directed actions), and perceptions (eudaimonic pleasure).

The purpose in life system posits that people high in future-oriented purpose in life possess a reasonable conception of how they wish to live. The perceived gap between their present state and ideal future life serves as a motivational force, driving people to reduce this discrepancy (Higgins, 1987). From the discrepancy between the present and ideal future life, people develop *self-concordant goals*, goals that are aligned with the person's core sense of self and identity (Sheldon, 2014; Sheldon & Elliot, 1999). Such goals provide structure for meaningful, *goal-directed actions*, actions that contribute to a personally valued outcome (Carver & Scheier, 1998; Emmons, 2003; Sheldon, 2014). Engaging with these actions stimulates eudaimonic pleasure, signifying a deep sense of flourishing, the fulfilment of one's potential, and the ongoing satisfaction of needs (Deci & Ryan, 2000; Huta, 2015; Sheldon & Elliot, 1999).

People low on future-oriented processes may feel lost and struggle to formulate meaningful goals. They may also lack intrinsic motivation and rely predominantly on external cues to set and pursue goals.

2.4 Purpose in Life as a System

The contribution of our proposed theoretical framework is not the introduction of a new construct per se but rather an integration of well-established motivational theories into a

comprehensive conceptualisation of purpose in life. Hence, our aim is not to prescribe the content of purpose in life, but to describe motivational processes through which purpose in life emerges. For example, individuals who experience contentment through independent pursuits and personal growth may develop purpose in life focussing on their own lives and the people closest to them. In contrast, those whose meaningful experiences have been shaped by commitments to idealistic or altruistic endeavours may find meaning and form a sense of identity and social role around contributing to the common good. Alternatively, it is possible that purpose in life can be fulfilled in relative solitude. For people who experience contentment, curiosity, and commitment toward things like art, music, intellectual pursuits, or nature, purpose in life may be sustained primarily through engagement in these personally meaningful domains.

This perspective builds on an emerging literature that frames purpose in life as a motivational system rather than a trait. It is therefore important that we distinguish our approach from three system-based approaches to purpose in life that we found to be prominent in the literature, namely McKnight and Kashdan's (2009) purpose in life theory, Schnell's (2009) hierarchical model of meaning, and Hill's (2023) Purpose as Trait, Habit, and State (PATHS) model.

McKnight and Kashdan (2009) define purpose in life as a "self-organising life aim that organises and stimulates goals, and manages behaviours, and provides a sense of meaning" (p. 242). McKnight and Kashdan's (2009) theory is one of the most well-known contemporary perspectives of purpose in life. However, we suggest that defining purpose in life as an aim makes it unclear how it differs from a higher-order goal. McKnight and Kashdan (2009) attempt to differentiate purpose in life and goals by stating that purpose in life "… is at the highest level of analysis and provides some degree of centrality in a person's identity" (p. 243). However, they do not specify the characteristics that allow a life aim to exert such a central influence, nor how it can adapt to life experience. We extend McKnight and Kashdan's work (2009) by viewing purpose in life as a flexible and evolving system that adjusts in response to personal growth and changing circumstances. We clarify the distinction between purpose in life and goals by positioning goals (specifically, self-concordant goals) as a component within a broader purpose in life system.

Schnell's (2009) hierarchical model of meaning uses the terms *meaning* and *purpose* interchangeably. In their model, Schnell (2009) depicts purpose or meaning as existing across five levels: perceptions, actions, goals, sources of meaning, and meaning in life, with the latter being the highest order, most complex, and most abstract level. A key feature of Schnell's (2009) model is internal coherence, reflected in consistency within (e.g. alignment between goals) and between (e.g. alignment between actions and goals) levels of meaning. While Schnell's (2009) model offers a broad structural framework, it could be argued that these levels are too general and vague to serve as an informative foundation for purpose in life. We therefore extend Schnell's (2009) work in our framework which identifies specific constructs and processes involved in purpose in life, which potentially offers greater clarity for operationalisation, measurement, and experimental manipulation.

The PATHS (Purpose as a Trait, Habit, and State) framework describes purpose in life development as a progression through three stages: first, purpose as a state, then as a habit, and ultimately as a trait (Hill et al., (2023). Once purpose becomes a trait, it then reinforces purpose as a state, creating a self-sustaining cycle (Hill et al., 2023). For example, a person who volunteers may first experience purpose in the transient pleasure in helping

others (state), with repeated engagement volunteering becomes regular intentional practice (habit), and eventually contributes to a self-concept organised around altruism and generosity (trait). In turn, the trait of altruistic or generous reinforces state purpose, through motivating ongoing engagement in volunteering. While this framework offers insight into present-oriented purpose in life processes, it notably omits future-oriented processes. Given the extensive literature linking purpose in life to meaningful goal engagement and sense of direction (Emmons, 2003; Lewis, 2020), this omission raises questions about whether the PATHS framework fully captures the mechanisms that drive purpose development. In contrast, we explicitly aim to incorporate future-oriented processes into our model, such as the conception of the ideal future, goal processes, goal-directed action, and eudaimonic pleasure. By integrating future-oriented processes, our model is more aligned with contemporary literature, accounting for how purpose in life promotes personal growth.

Based on our framework, purpose in life shares some overlap with wellbeing however, it remains conceptually and empirically distinct. Wellbeing represents an emotional evaluative state (Jarden & Roache, 2023), whereas purpose in life is an ongoing, internal, motivational system (Lewis, 2020). Moreover, purpose in life does not always align with positive wellbeing and has been empirically shown to coincide with increased psychological symptoms, such as subjective stress (Sutin et al., 2024). This may occur because having a meaningful sense of direction, at least in the short term, can be overly taxing on cognitive and emotional resources and generate increased distress (e.g., quitting a stable but unfulfilling job to pursue a more meaningful career). Purpose in life and wellbeing are therefore not conceptually interchangeable and should be measured separately.

3 The Multidimensional Measure of Purpose (MMOP)

Based on our integrative framework of purpose in life, we developed the Multidimensional Measure of Purpose in Life (MMOP). The MMOP is a 12-item questionnaire designed to measure three constructs that encapsulate our purpose in life model. Present-oriented purpose in life (P-PIL) reflects ongoing engagement in life through reciprocal interactions between feelings of contentment, exploration, commitment, and the development of one's identity and social role. As such, P-PIL is captures how purpose is experienced and enacted in the present through meaningful daily engagements. Future-oriented purpose in life (F-PIL) reflects organisation of motivational processes around conceptions the desired life, which structure self-concordant goals, goal directed actions, and eudaimonic pleasure. F-PIL therefore provides purpose through a valued sense of direction. Purpose congruence across time (PCAT) reflects a connection between past, present, and future orientations. PCAT is therefore the sense of continuity across life and the formation of a cohesive life story. Together these components represent complementary processes within the purpose in life system (as seen in Fig. 1) such that purpose in life emerge from their dynamic integration, rather than any single component in isolation.

The MMOP instructs participants to score items based on their experiences on a typical day to anchor responses to habitual patterns of thinking, feeling, and behaviour. This approach is to measure a more stable, general sense of purpose in life rather than day-to-day transient feelings of purpose. Respondents rate each item using a 7-point Likert scale ranging from 1 (*absolutely untrue*) to 7 (*absolutely true*). Subscales for each dimension are

summed to produce a composite purpose in life score, with higher scores indicating a higher level of purpose in life. No items were reverse scored as previous literature has shown this to be ineffective at reducing response biases and negatively affects the psychometric properties of the scale (van Sonderen et al., 2013; Woods, 2006). A 7-point scale was chosen based on Kyriazos and Stalikas' (2018) recommendations for a balance between sensitivity and reduced risks of random error. All scale points are labelled to help respondents better understand their options, leading to more reliable and accurate responses (Kyriazos & Stalikos, 2018).

Each item of the MMOP was designed to correspond with a specific interaction in the Purpose in Life model (e.g., "*discovering new things feels satisfying*" targets the process linking hedonic pleasure and curious exploration). Most processes within the purpose in life model are represented in the MMOP items with only three exceptions: the processes linking (1) motivated commitment and identity and social role formation, (2) goal-directed actions and eudaimonic pleasure, and (3) history of met and unmet needs and hedonic pleasure. These exceptions were excluded during the item selection phase of the MMOP (detailed in Study 1) due to highly leptokurtic and negatively skewed score distributions, rendering these items unsuitable for inclusion in this version of the MMOP.

Items within each MMOP dimension were worded simply to accurately convey the intended construct. Given the experiential similarities between hedonic pleasure and eudaimonic pleasure, we used broad terms (e.g., content, satisfaction, positive) to refer to these constructs, avoiding specific states such as happiness and joy (Huta, 2015). Hedonia and eudaimonia were instead differentiated by their temporal qualities, with hedonic items focusing on present-oriented feelings of pleasure and eudaimonia items focusing on future-oriented feelings of pleasure (Huta, 2015).

Based on qualitative feedback gathered from the pilot phase of the MMOP, we refrained from using the term *goals* in the MMOP due to respondents generally believing that goals had to be fixed, objective, and/or a measurable aim. As we wished to address goal-related processes more broadly in the MMOP, we phrased items to ask about how the respondent desires to live their life. Additionally, psychological needs are not explicitly mentioned in the MMOP due to its value-laden nature (Vansteenkiste & Ryan, 2013). Instead, we ask about past experiences, eliciting responses that involve emotionally salient histories which likely involve experiences of met and unmet psychological needs (Vansteenkiste & Ryan, 2013).

Further considerations have been made regarding the tone of items to accurately target the key aspects of the assessed constructs. For instance, all items of the P-PIL subscale were process-oriented and worded to capture the ongoing, evolving nature of exploration and development. Hence, we preferred the wording "*I dedicate myself to the things that are meaningful*" over "*I am dedicated to the things that are meaningful*" as the former conveys a proactive and continuous approach regarding *how* an individual dedicates themselves, while the latter represents a static state, an outcome of dedication (Hill et al., 2023; Lewis, 2020).

The F-PIL subscale comprises a combination of process-orientated items (e.g., *I do things that are consistent with how I want to live my life)* to highlight the aspects of purpose in life concerned with striving, as well as evaluative items (e.g., *"I feel content with how my life is going")* to target the sense of progress toward meaningful life aims. Items of the PCAT subscale are structured similarly to scales measuring narrative identity and temporal congruence (e.g., awareness of narrative identity questionnaire; Hallford, 2017) and reflect

processes connecting past, present, and future orientations (e.g., "*Thinking about who I am today helps me to figure out how I want to live in the future"*).

The development of the MMOP consisted of two psychometric studies. Study 1 describes the item selection process and the initial examination of the MMOP's factor structure and internal consistency. Study 2 aims to replicate Study 1 to examine the factor structure and internal consistency using a different sample and to test discriminant, convergent, and concurrent validity.

4 Study 1

4.1 Participants and Procedure

A total sample of 238 participated in Study 1. Of these, 10 cases were deleted as they were missing more than 5% of data, which we considered a conservative threshold to minimise excessive data replacement. One case was deleted for providing the same answer to all items, suggesting inattentive responding. The remaining two missing data points were replaced with expectation maximisation (Kline, 2023).

The final sample comprised 227 English-speaking adults (Female = 66.5%; *M* age = 30.89, SD = 12.01) residing in Australia, meeting sample size recommendations for factor analyses of 10 participants per item (Boateng et al., 2018). Participants were recruited via two primary methods: advertising on social media platforms and a university's student participation pool, with the latter offering course credit for their involvement. Demographic details for student and community samples are provided in Table 1 separately and combined. An independent samples t-test showed no significant difference ($p > .05$) between the MMOP scores of student and community samples; therefore, analyses were conducted using the total sample.

The MMOP items were administered using an anonymous online questionnaire within an 'overinclusive' pool of 96 items. We designed 32 items to measure P-PIL(e.g., *When something inspires me*, *I find ways to pursue it further)*, 32 items to measure F-PIL (e.g., *I have a good idea of how I want to live my life and the steps to making this a reality*), and 32 items to measure PCAT (e.g., *The more I grow in life*, *the more I can appreciate the efforts I have made in the past)*. The initial item pool was generated by the research team (including a practising clinical psychologist) and included items that were adapted from existing measures that appear to target these three respective dimensions (e.g., Multidimensional Existential Meaning Scale; George & Park, 2017). Subsequently, the items were piloted with a sample of 21 psychologists and community members to receive qualitative feedback regarding item clarity, relevance, and acceptability. Feedback comments guided modifications to the item pool to improve the wording and clarity of questions, generally, these changes were made to help participants think more broadly about purpose in life (e.g., replacing the word *goals* with phrases such as *how I want to live my life*). All 96 items and relevant statistics have been made available at [link to be provided upon manuscript approval].

Table 1 Demographic Information for Study 1

	Student Sample ($n=169$)	Community Sample ($n=58$)	Combined ($n=227$)
Age			
Mean	28.54	37.84	30.89
Range	18 to 61	20 to 70	18 to 70
SD	10.16	14.288	12.01
Prefer not to answer or invalid response, n (%)	3 (1.8%)	2 (3.4%)	5 (2.2%)
Gender, n (%)			
Female	125 (74.0%)	22 (37.9%)	147 (66.5%)
Male	37 (21.9%)	32 (55.2%)	69 (30.3%)
Non-binary	2 (1.2%)	1 (1.7%	3 (1.3%)
Gender Fluid	1 (0.6%)	-	2 (0.9%)
Transgender	1 (0.6%)	1 (1.7%)	1 (0.4%)
Prefer not to answer or invalid response	3 (1.8%)	2 (3.4%)	5 (2.2%)
Ethnicity, n (%)*			
Australian Aboriginal or Torres Strait Islander	12 (7.1%)	1 (1.7%)	13 (5.7%)
Asian	23 (13.6%)	9 (15.5%)	32 (14.1%)
Black or African	8 (4.7%)	1 (1.7%)	9 (4.0%)
Hispanic, Latino, Spanish Origin	5 (3.0%)	2 (3.4%)	7 (3.1%)
Māori or other Pacific Islander	4 (2.4%)	2 (3.4%)	6 (2.6%)
Middle Eastern or North African	5 (3.0%)	4 (6.9%)	9 (4.0%)
White	130 (76.9%)	47 (81.0%)	177 (78.0%)
Prefer not to answer or invalid response	3 (1.8%)	1 (1.7%)	4 (2.2%)
Highest level of education, n (%)			
High school diploma or equivalent	-	7 (12.1%)	7 (3.1%)
Vocational training	-	1 (1.7%)	1 (0.9%)
Certificate or Diploma	-	11 (19.0%)	11 (4.9%)
Bachelor's degree	153 (90.53%)	21 (36.2%)	173 (76.2%)
Some postgraduate work	13 (7.7%)	4 (6.9%)	17 (7.5%)
Master's Degree	2 (1.2%)	10 (17.2%)	12 (5.3%)
Doctorate Degree	-	3 (5.2%)	3 (1.3%)
Prefer not to answer or invalid response	1 (0.6%)	1 (1.7%)	2 (0.9%)

4.2 Item Selection

To explore and examine the underlying factor structure of the pool of 96 items, a series of preliminary exploratory factor analyses (EFA) were conducted on items of each of the three proposed MMOP dimensions. EFA outputs for the final MMOP items are displayed in Supplementary Materials 1.

Based on these analyses, we selected the 12 items to carry forward for Confirmatory Factor Analysis (CFA). The MMOP item selection process aimed to achieve even representations for each of its three dimensions, with at least three items per dimension (Carpenter, 2018). Moreover, items were excluded if they did not load significantly (factor loading $\geq$ 0.40) onto the intended factor or cross-loaded onto another dimension. If multiple items seemed to measure the same construct, either semantically or quantitatively, with a high

Pearson's correlation (>0.90), the item that appeared most accurate and parsimonious was kept for inclusion in the MMOP. In this paper, we report only on the results of the 12 items retained for the MMOP.

4.3 Materials

4.3.1 Demographic Questions

Demographic variables included information regarding gender identity, age, education, and ethnicity.

4.3.2 The Multidimensional Measure of Purpose in Life

The MMOP is a 12-item measure of purpose in life (as described above). The final version of the MMOP is presented in Supplementary Materials 2.

4.4 Analysis

4.4.1 Analysis Software

CFAs were conducted using the Lavaan package (Rosseel, 2012) in R studio version 4.4.0. All other analyses were conducted using SPSS 24.

4.4.2 Descriptive Statistics and Missing Data

Means and standard deviations were computed for the total MMOP score and each of the three subscales.

4.4.3 Internal Consistency Reliability

To assess internal consistency reliability, we calculated Cronbach's alpha coefficient. Reliability coefficients ≥ 0.60 were considered acceptable but modest (Hair et al., 2009),≥0.70 acceptable, ≥ 0.80 were considered good, and ≥. 90 were considered excellent (Groth-Marnat & Wright, 2016).

4.4.4 Factor Analysis

CFAs were conducted using both maximum likelihood expectation (ML) and Bentler and Yuan's (1999) robust maximum likelihood (MLR). MLR provides more accurate estimates than ML with data that deviates from a normal distribution (Bentler & Yuan, 1999; Curran et al., 1996). Three theoretically informed models were tested (see Fig. 2).

(1) Model 1 is a one-factor model where all 12 items of the MMOP load onto a single purpose in life factor.

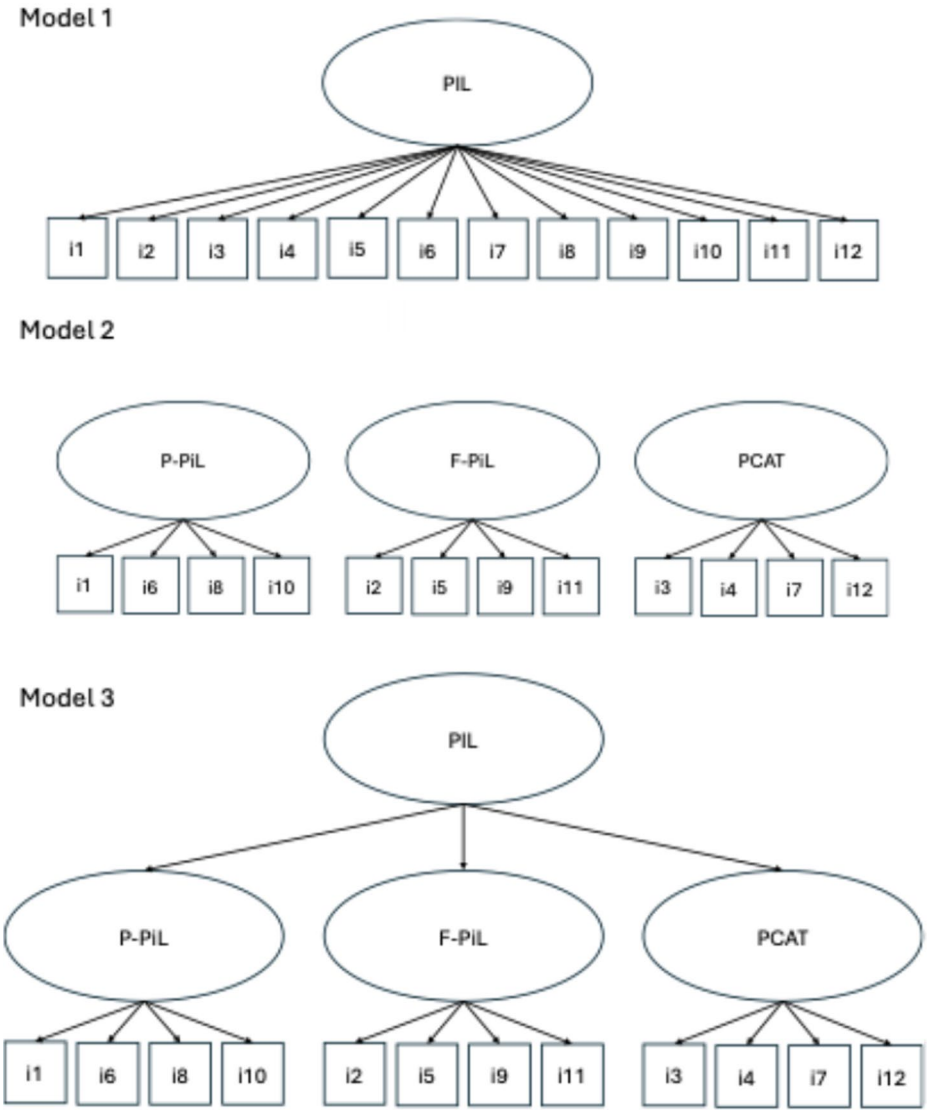


Fig. 2 Confirmatory factor analysis models assessed in Study 1. Note. P-PIL = Present oriented Purpose in Life, F-PIL = Future oriented Purpose in Life, PCAT = Purpose congruence across time, i = item number

(2) Model 2 is a three-factor uncorrelated model where all items load onto one of three primary factors: P-PIL (items 1, 6, 8, 10), F-PIL (items 2, 5, 9, 11), and PCAT (items 3, 12, 4, 7).
(3) Model 3 is a higher-order model with MMOP items loading onto three primary factors (P-PIL, F-PIL, and PCAT) which then load onto a single higher-order factor, purpose in life.[1]

4.4.5 Goodness of Fit

To evaluate the model fit we examined factor loadings and the intercorrelations within each model. Additionally, we also calculated five fit indices: $\chi 2$ statistic (or scaled $\chi 2$ for MLR), the comparative fit index (CFI), Tucker Lewis Index (TLI), root mean square error of approximation (RMSEA), and standardised root mean residual (SRMR). As $\chi 2$ is highly influenced by non-normal distributions and sample size, we referred primarily to CFI, TLI, RMSEA, and SRMR to judge the model fit (Kline, 2023). Models were considered acceptable if CFI and TLI values were ≥ 0.90 and RMSEA and SRMR were ≤ 0.08 (Hu & Bentler, 1999; Kline, 2023; Marsh et al., 2004).

4.5 Results and Discussion

4.5.1 Descriptive Statistics and Internal Consistency Reliability

Descriptive and Internal consistency reliability statistics for the proposed factors of MMOP are presented in Table 2.

Cronbach's alpha coefficients for the Study 1 sample demonstrated good internal consistency across all three MMOP subscales and as a total score, with internal consistency coefficients ranging from 0.80 to 0.87.

4.5.2 Factor Analysis

Model 1 showed a poor fit with all fit indices not meeting thresholds for ML ($\chi 2 = 543.774$, df = 54, $p < .001$, CFI = 0.545, TLI = 0.444, SRMR = 0.126, RMSEA = 0.175) or RML ($\chi 2 = 431.304$, df = 54, $p < .001$, CFI = 0.617, TLI = 0.532, SRMR = 0.126, RMSEA = 0.197). Model 2 showed an improved fit to data, though did not meet acceptable fit for any of the fit indices for both ML ($\chi 2 = 244.381$, df = 24, $p < .001$, CFI = 0.866, TLI = 0.832, SRMR =

Table 2 Descriptive Statistics, alpha coefficient (α) and Pearson's correlations coefficients between dimensions of the Multidimensional Measure of Purpose in Life (MMOP) in Study 1

	Mean	SD	α	PREF	PEXP	PCAT
P-PIL	22.6	3.77	0.8	-	0.446	0.476
F-PIL	21.46	4.5	0.84	-	-	0.498
PCAT	22.17	4.71	0.87	-	-	-
Total Purpose in life score	66.55	10.2	0.87			

[1] Given the large inter-factorial correlations, bifactorial models were also examined. While a bifactor model was estimable for Study 1, it failed to converge in Study 2 due to extremely high item intercorrelations resulting in model instability. As a stable bifactor structure could not be supported, and the higher-order model demonstrated adequate fit in both studies, the bifactor solutions was not retained.

0.235, RMSEA = 0.096) and RML ($\chi2 = 168.114$, df = 24, $p<.001$, CFI = 0.862, TLI = 0.837, SRMR = 0.235, RMSEA = 0.116). In contrast, CFA with Model 3 showed further improved results with ML results adequately meeting all fit indices ($\chi2 = 131.195$, df = 51, $p<.001$ CFI = 0.947, TLI = 0.932, SRMR = 0.054, RMSEA = 0.062), suggesting that the MMOP captures a multidimensional construct that includes a high-order factor. Robust estimators confirmed these improvements with all RML fit indices showing good fit to data (Scaled $\chi2 = 94.798$, df = 51, $p<.001$, CFI = 0.951, TLI = 0.937, SRMR = 0.054, RMSEA = 0.072). Standardised factor loadings for Model 3 are presented in Table 3.

As seen in Table 3, all items loaded well onto the intended factor (between 0.628 and 0.855). Item uniqueness ranged between 0.269 and 0.612 suggesting that the items had sufficient shared variance with the designated factors, while avoiding excessive redundancy. Model 3 therefore, demonstrates adequate model fit, strong primary loadings, and acceptable uniqueness values, indicating good item-factor discrimination within a higher order structure. Detailed outputs have been made available in Supplementary Materials 3.

5 Study 2

5.1 Participants and Procedure

To replicate and extend the results of Study 1, the MMOP was administered to a new sample comprising 286 participants (Female = 50.18%) with a mean age of 34.80 years

Table 3 Standardised Factor Loadings from Confirmatory Factor Analysis of the Multidimensional Measure of Purpose in Life in Study 1

Item number and Scale	Factor Loadings
PREF	
1. I dedicate myself to the things that feel meaningful	0.766
6. When something inspires me, I find ways to pursue it further	0.811
8. Discovering new things feels satisfying	0.623
10. When something feels meaningful to me, I seek to understand why	0.628
F-PIL	
2. I do things that are consistent with how I want to live my life	0.646
5. I have a good idea of how I want to live my life and the steps to making this a reality	0.693
9. I feel content with how my life is going	0.823
11. I feel positive about the progress I have made in life	0.855
PCAT	
3. Thinking about who I am today helps me to figure out how I want to live in the future	0.763
4. The more I grow in life, the more I can appreciate the efforts I have made in the past	0.809
7. Looking back at the things I learned in the past has helped me to form an idea of my ideal future life	0.822
12. Knowing more about my past has provided clarity on who I am now	0.795
Total MMOP score	
PREF	0.759
PEXP	0.691
PCAT	0.678

Note. P-PIL = Present-oriented Purpose in Life, F-PIL = Future-oriented Purpose in Life, PCAT = Purpose Congruence Across Time

(SD = 12.00). This sample exceeded the minimum sample size ($n = 84$) determined by an a priori power analysis in G*Power (Faul et al., 2007) for detecting medium-sized correlations of (r =.30). The power calculation is based on Cohen's (1992) recommendation to detect medium effects at a power of 0.80 with an alpha of 0.05 when conducting behavioural science research studies. The total sample size also met the criteria for conducting factor analysis based on the criteria of 10 participants per item (Boateng et al., 2018). As with Study 1, participants were English-speaking Australian residents recruited using an online panel service (Prolific). Further information regarding gender, ethnicity, and education is presented in Table 4.

5.2 Materials

In addition to the demographic questions and the MMOP. This study also included several measures to assess the MMOP's concurrent, convergent, and discriminant validity Following the MMOP, the additional scales were administered in the order presented and described below.

5.2.1 Life Engagement Test (LET; Scheier et al., 2006)

The LET is a well-established unidimensional measure of purpose in life. The LET comprises 6 items (e.g., "*most of what I do is trivial to me"*), with response options ranging from 1 = *strongly disagree* to 7 = *strongly agree*. The LET has shown good reliability and validity in past studies (Pearson et al., 2013; Scheier et al., 2006). The LET showed a Cronbach's alpha coefficient of 0.88 in the current study.

5.2.2 The Purpose in Life Scale- short form (PILS-SF; Schulenberg et al., 2011)

The PIL-SF is a self-report measure of purpose in life. The PIL-SF comprises 4 items, each rated on a scale ranging from 1 (e.g., *I have discovered no mission or Purpose in life)* to 7 (e.g., *I have discovered a satisfying life Purpose*). Scores on the PIL-SF range from 4 to 28, with higher scores indicating a greater sense of Purpose in life. The PILS-SF has shown good validity and reliability (Schulenberg et al., 2011; Weber et al., 2020). In the present study, the PILS-SF showed an internal consistency coefficient of 0.88.

5.2.3 Curiosity and Exploration Inventory II (CEI-II; Kashdan et al., 2009)

The CEI-II is a self-report measure to assess trait curiosity. The CEI-II consists of 10 items rated on a 5-point Likert Scale from 1 = *very slightly or not at all* to 5 = *extremely*. Items of the CEI-II comprise two subscales: *Stretching*, the motivation to seek out novel experiences and embracing (e.g., *I actively seek as much information as I can in new situations*, and *embracing*, the willingness to embrace unpredictability and uncertainty (e.g., *I am the type of person who enjoys the uncertainty of everyday life*). Subscale scores are attained by summing the items of the respective scale. Subscale scores can also be combined to produce a total CEI-II score, with higher scores indicating a greater proclivity toward curiosity and exploration. The CEI-II has shown good validity and reliability in past research (Kashdan et

Table 4 Demographic Information for Study 2

	Total sample (n = 286)
Age	
Mean	34.8
Range	18 to 83
SD	12
Prefer not to answer or invalid response, n (%)	3 (1.0%)
Gender, n (%)	
Female	142 (50.18%)
Male	137 (48.41%)
Non-binary	2 (0.70%)
Agender	1 (0.35%)
Transgender	1 (0.35%)
Prefer not to answer or invalid response	3 (1.0%)
Ethnicity, n (%)*	
Australian Aboriginal or Torres Strait Islander	18 (6.2%)
Asian	86 (30.1%)
Black or African	4 (1.4%)
Hispanic, Latino, Spanish Origin	0 (0.0%)
Māori or other pacific islander	8 (2.8%)
Middle Eastern or North African	4 (1.4%)
White	188 (65.8%)
Prefer not to answer or invalid response	0 (0.0%)
Highest level of education, n (%)	
Attended high school	6 (2.8%)
High school diploma or equivalent	30 (10.5%)
Vocational training	4 (1.4%)
Attended some Tafe/College/University	24 (8.4%)
Certificate or Diploma	35 (12.3%)
Bachelor's degree	112 (39.2%)
Some postgraduate work	21 (7.3%)
Master's Degree	47 (16.4%)
Doctorate Degree	7 (2.4%)
Prefer not to answer or invalid response	0 (0.0%)

Note. *Participants were permitted to select more than one answer for this question

al., 2009). The alpha coefficients for the CEI-II in the current study were 0.75 for the stretching subscale, 0.87 for the embracing subscale, and 0.89 for the total CEI-II.

5.2.4 Goal Adjustment Scale (GAS; Wrosch et al., 2003)

The GAS is a 10-item measure designed to assess the ability to adjust to setbacks in the pursuit of meaningful goals. Items are rated on a scale from 1 = *strongly disagree* to 5 = *strongly agree.* The GAS comprises two distinct dimensions measured with 5 items each, *Goal disengagement* is the ability to effectively abandon blocked goals (e.g., *It's easy for me to reduce my effort towards the goal*). The second dimension is *goal reengagement*, the ability to effectively reengage with an alternative, meaningful goal (e.g., *I tell myself I have other meaningful goals to pursue*). Subscale totals are attained by summing items of the respec-

tive scale. The GAS has demonstrated good validity and reliability (Wrosch et al., 2003). In the present study, the goal disengagement scale showed an alpha coefficient of 0.86, while the goal reengagement scale showed an alpha coefficient of 0.79.

5.2.5 The Scale of Positive and Negative Experience (SPANE; Diener et al., 2009)

The SPANE is a 12-item, self-report questionnaire used to assess emotionality through two subscales: pleasant emotions (e.g., *happy*) and unpleasant emotions (e.g., *sad*). Participants rate 12 emotion items on a 5-point scale ranging from 1 = *Very rarely/Never* to 5 = *Very Often/Always*, with subscale scores summed separately. The SPANE has demonstrated good validity and reliability (Diener et al., 2009). In the current study, the SPANE showed a good internal consistency coefficient of 0.92 for the positive emotion subscale and 0.85 for the negative emotion subscale.

5.2.6 Awareness of Narrative Identity Questionnaire (ANIQ; Hallford & Mellor, 2017)

The ANIQ is a 20-item self-report measure to assess how coherent a person perceives their memories. ANIQ items are rated on a scale from 0 = *completely disagree* to 10 = *completely agree* across four subscales: awareness, temporal coherence, causal coherence, and thematic coherence. The subscales can be summed to produce a total score. The ANIQ has demonstrated good reliability and validity (Hallford & Mellor, 2017). The current study showed good internal consistencies among all subscales of the ANIQ with Cronbach's alphas between 0.92 and 0.96 for all subscales and 0.96 for the total ANIQ score.

5.2.7 Adult Hope Scale (AHS; Snyder, 1991)

The AHS is a self-report measure used to assess hope. The AHS consists of 12 items, with each item rated on an 8-point Likert scale ranging from 1 = *definitely false* to 8 = *definitely true*. The AHS comprises two subscales: *Agency*, one's determination to achieve goals (e.g., *I energetically pursue my goals*), and *Pathways*, one's plans to achieve desirable outcomes (e.g., *There are lots of ways around any problem*). Subscale scores can also be summed to produce an overall measure of hope. The AHS has demonstrated good reliability and validity in past studies (Snyder, 1991). The present research showed a Cronbach's alpha coefficient of 0.85 for the Agency subscale, 0.80 for the Pathway subscale, and 0.88 for the overall AHS.

5.2.8 General Anxiety Disorder – 7 (GAD-7; Spitzer et al., 2006)

The GAD-7 measures respondents' experience of anxiety over the previous two weeks. The GAD-7 consists of seven items (e.g., *worrying too much about different things*) rated on a 4-point scale from 0 = *not at all* to 4 = *nearly every day*. Scores on the GAD-7 are summed to provide a total score. Established cut-off criteria indicate four levels of anxiety: *minimal anxiety* (0 to 4), *mild anxiety* (5 to 9), *moderate anxiety* (10 to 14), and *severe anxiety* (15 to 21). The GAD-7 has demonstrated good validity and reliability in clinical and non-clinical samples (Plummer et al., 2016; Spitzer et al., 2006). The present study showed an alpha coefficient of 0.91.

5.2.9 Patient Health Questionnaire–9 (PHQ-9; Kroenke et al., 2001)

The PHQ-9 is a self-report measure used to assess depressive symptoms over the previous two weeks. The PHQ-9 consists of nine items (e.g., *feeling down*, *depressed*, *or hopeless*). Each item is rated on a 4-point scale from 0=*not at all* to *nearly* 4=*every day*. This score can then be categorised into one of five severity levels: *minimal* (0 to 4), *mild* (5 to 9), *moderate* (10 to 14), *moderately severe* (15 to 19), and *severe* (20 to 27). The PHQ-9 has demonstrated good validity and reliability in clinical and non-clinical samples (Löwe et al., 2004; Martin et al., 2006). The PHQ-9 demonstrated a Cronbach's correlation coefficient of 0.89 in the current study.

5.3 Analysis

5.3.1 Descriptive Statistics, Factor Analysis, Internal Consistency Reliability

We calculated descriptive statistics, factor structure and internal consistency reliability using the same process as described in Study 1.

5.3.2 Concurrent Validity

To assess concurrent validity, Pearson's correlation coefficients were conducted to examine whether MMOP's associations with other measures of purpose in life, the PILS-SF and LET.

5.3.3 Convergent Validity

Pearson's correlations were used to assess relationships between the MMOP and well-established measures of positive affect, goal adjustment, curiosity and engagement, and hope. We expect the positive affect dimension of the SPANE to correlate significantly and positively with the MMOP. Positive correlations between the MMOP and CEI-II reflect purpose as a process that fosters learning and exploration, as indicated by (Kashdan and McKnight, 2009). Positive relationships between the MMOP and the goal reengagement and hope align with the idea that purpose in life promotes adaptive goal pursuit (Lewis; 2020). Furthermore, positive relationships between the MMOP and ANIQ explore the temporal dimension of purpose in life, linking it with a congruence between past and current orientations toward meaningful future direction (Stillman & Baumeister, 2009).

5.3.4 Divergent Validity

Divergent validity was evaluated by examining Pearson's correlations between the MMOP and the GAD-7, PHQ-9, and SPANE's negative emotion dimension. We selected these measures to ensure we capture purpose in life as a unique aspect of well-being and conceptually distinct from anxiety, depression, and unpleasant emotional states.

5.4 Results and Discussion

5.4.1 Descriptive Statistics and Internal Consistency Reliability

In the current study, the P-PIL subscale had a mean of 22.20 (*SD* = 3.27). The F-PIL.

subscale had a mean of 20.05 (SD = 4.60). The PCAT mean was 21.73 (SD = 3.80), and the mean for the total MMOP scale is 63.98 (SD = 10.08). Cronbach's alpha coefficient for the P-PIL subscale was 0.68, which is lower in the current study compared to Study 1. Internal consistency remained strong for F-PIL, PCAT, and overall MMOP scales, which were 0.84, 0.82, and 0.89, respectively.

5.4.2 Factor Structure

As in Study 1, the results of the current study showed that a single factor model, Model 1, was a poor fit to data using ML ($\chi2 = 325.898$, df = 54, $p<.001$, CFI = 0.828, TLI = 0.790, SRMR = 0.073, RMSEA = 0.116) or RML ($\chi2 = 260.427$, df = 54, $p<.001$, CFI = 0.830, TLI = 0.792, SRMR = 0.073, RMSEA = 0.130). In this study, Model 2 showed a poorer fit to data compared to Model 1 with both ML ($\chi2 = 544.535$, df = 54, $p<.001$, CFI = 0.678, TLI = 0.606, SRMR = 0.313, RMSEA = 0.159) and RML ($\chi2 = 441.205$, df = 54, $p<.001$, CFI = 0.685, TLI = 0.615, SRMR = 0.313, RMSEA = 0.177). Model 3 showed the best fit across the three models tested, as shown in ML ($\chi2 = 212.959$, df = 51, $p<.001$, CFI = 0.895, TLI = 0.864, SRMR = 0.068, RMSEA = 0.093) and RML ($\chi2 = 177.514$, df = 51, $p<.001$, CFI = 0.900, TLI = 0.871, SRMR = 0.068, RMSEA = 0.102) indices. While some Model 3 indices were near but did not meet threshold criteria, the overall pattern loadings suggest reasonable fit for this higher order model. Standardised item loadings ranged from 0.422 to 0.874, while uniqueness estimates ranged from 0.236 to 0.822. Supporting item-factor discrimination, these statistics indicate that loaded strongly onto the designated factor and avoided excessive redundancy. Relevant item loadings are presented in Table 5. Detailed outputs including uniqueness estimates are shown in Supplementary Materials 4.

5.4.3 Concurrent Validity

Supporting the concurrent validity of the MMOP, the LET and PIL-SF showed significant, positive correlations with all subscales of the MMOP as well as the total MMOP score. The LET and PIL-SF showed the strongest relationships with the F-PIL subscale (r =.69 and 0.76, respectively), suggesting that these measures are associated purpose in life with a focus on a future orientation.

5.4.4 Convergent Validity

Table 6 displays the relationships between MMOP subscales and theoretically related variables used in the current study. Curious exploration, goal reengagement, hope, awareness of narrative identity, and positive affect significantly correlated with the MMOP subscales. Notably, the temporal dimension of ANIQ showed high correlations with all MMOP subscales (r =.75 to 0.93), suggesting a conceptual overlap between temporal congruence across the MMOP dimensions. Moreover, the agency scale of the AHS showed large, significant

Table 5 Standardised Factor Loadings from Confirmatory Factor Analysis of the Multidimensional Measure of Purpose in Life in Study 2

Item number and Scale	Factor Loadings
P-PIL	
1. I dedicate myself to the things that feel meaningful	0.632
6. When something inspires me, I find ways to pursue it further	0.729
8. Discovering new things feels satisfying	0.422
10. When something feels meaningful to me, I seek to understand why	0.592
F-PIL	
2. I do things that are consistent with how I want to live my life	0.631
5. I have a good idea of how I want to live my life and the steps to making this a reality	0.748
9. I feel content with how my life is going	0.801
11. I feel positive about the progress I have made in life	0.874
PCAT	
3. Thinking about who I am today helps me to figure out how I want to live in the future	0.746
4. The more I grow in life, the more I can appreciate the efforts I have made in the past	0.758
7. Looking back at the things I learned in the past has helped me to form an idea of my ideal future life	0.71
12. Knowing more about my past has provided clarity on who I am now	0.679
Total MMOP	
PREF	0.879
PEXP	0.763
PCAT	1.008

relationships with F-PIL and total MMOP subscale (0.78 and 0.71, respectively), reflecting the link between psychological needs and the processes involved in goal pursuit (Ryan & Deci, 2000).

5.4.5 Divergent Validity

As seen in Table 6, negative affect, goal disengagement, anxiety and depression demonstrated mostly significant negative correlations with the MMOP subscales and total MMOP score. Although the relationships between future purpose in life and goal disengagement and between present purpose in life and anxiety were negative, they did not reach statistical significance.

6 General Discussion

The present paper introduces a novel theoretical framework of purpose in life, conceptualising it as a dynamic system that integrates multiple motivational constructs. The purpose in life system offers a holistic perspective, incorporating the past, present, and future orientations of purpose in life, thereby capturing its evolving influence over time and context. By

Table 6 Pearson's Correlations between the MMOP, and measures of concurrent, convergent, and divergent validity

	a	1	2	3	4	5	6	7	8	9	10	11
1. P-PIL	0.68	-										
2. F-PIL	0.84	0.52**	-									
3. PCAT	0.82	0.67**	0.65**	-								
4. Total MMOP	0.89	0.82**	0.87**	0.89**	-							
5. LET	0.88	0.44**	0.69**	0.54**	0.66**	-						
6. PIL-SF	0.88	0.48**	0.76**	0.53**	0.71**	0.74**	-					
7. CEI II Stretching	0.75	0.47**	0.33**	0.45**	0.48**	0.34**	0.43**	-				
8. CEI II Embracing	0.87	0.37**	0.23**	0.28**	0.33**	0.28**	0.38**	0.70**	-			
9. CEI II Total	0.89	0.46**	0.32**	0.40**	0.44**	0.35**	0.45**	0.89**	0.94**	-		
10. GAS GD	0.86	−0.24**	−0.11	−0.19**	−0.20**	−0.16*	−0.23**	−0.16*	−0.08	−0.14†	.	
11. GAS GR	0.79	0.32**	0.36**	0.36**	0.40**	0.39**	0.31**	0.26**	0.29**	0.31**	0.1	-
12. SPANE Positive	0.92	0.36**	0.71**	0.46**	0.61**	0.64**	0.65**	0.37**	35**	0.40**	−0.01	0.35**
13. SPANE Negative	0.85	−0.19*	−0.53**	−0.24**	−0.40**	−0.45**	−0.48**	−0.20**	−0.16*	−0.20**	−0.03	−0.17*
14. ANIQ Awareness	0.93	0.41**	0.34**	0.59**	51**	0.36**	0.32**	0.35**	0.28**	0.34**	−0.15	0.24**
15. ANIQ Temporal	0.96	0.88**	0.75**	0.79**	0.93**	0.58**	0.62**	0.49**	0.37**	0.46**	−0.17*	0.37**
16. ANIQ Causal	0.92	0.21**	0.13†	0.23**	0.21**	0.18*	0.01	0.21**	0.11	0.15*	−0.11	0.06
17. ANIQ Thematic	0.95	0.41**	0.30**	0.47**	0.44**	0.28**	0.23**	0.35**	0.25**	0.32**	−0.22**	0.20**
18. ANIQ Total	0.96	0.42**	0.31**	0.52**	0.48**	0.34**	0.28**	0.36**	0.25**	0.32**	−0.20**	0.20**
19. AHS Agency	0.85	0.41**	0.78**	0.58**	0.71**	0.67**	0.76**	0.45**	0.38**	0.46**	−0.19**	0.34**
20. AHS Pathway	0.8	0.44**	0.48**	0.47**	0.54**	0.46**	0.49**	0.54**	0.44**	0.53**	−0.14	0.29**
21. AHS Total	0.88	0.43**	0.53**	55**	0.59**	0.44**	0.49**	0.38**	0.29**	0.37**	−0.17*	0.30**
22. GAD 7	0.91	−0.10	−0.43**	−0.16*	−0.29**	−0.41**	−0.44**	−0.14†	−0.12†	−0.15†	−0.02	−0.14†
23. PHQ 9	0.89	−0.16*	−0.54**	−0.26**	−0.40**	−0.56**	−0.54**	−0.15†	−0.12†	−0.15†	−0.00	−0.17*
	12	13	14	15	16	17	18	19	20	21	22	23
1. PREF												
2. PEXP												
3. PCAT												
4. Total MMOP												

Table 6 (continued)

	a	1	2	3	4	5	6	7	8	9	10	11
5. LET												
6. PIL-SF												
7. CEI II Stretching												
8. CEI II Embracing												
9. CEI II Total												
10. GAS GD												
11. GAS GR												
12. SPANE Positive	-											
13. SPANE Negative	−0.69**	-										
14. ANIQ Awareness	0.31**	−0.01	-									
15. ANIQ Temporal	57**	−0.38**	0.44**	-								
16. ANIQ Causal	0.16*	−0.12	0.07	0.40**	-							
17. ANIQ Thematic	0.22**	−0.05	0.23**	0.41**	0.48**	.-						
18. ANIQ Total	0.29**	−0.11	0.81**	0.44**	0.79**	0.86**	-					
19. AHS Agency	0.68**	−0.53**	0.41**	0.62**	0.21**	0.35**	0.40**	-				
20. AHS Pathway	0.51**	−0.35**	0.38**	0.50**	0.29**	0.38**	0.43**	0.69**	-			
21. AHS Total	0.40**	−0.16*	0.48**	0.52**	0.21**	0.44**	0.46**	0.73**	0.74**	-		
22. GAD 7	−0.57**	0.70**	0.12	−0.27**	−0.01	0.03	−0.05	−0.45**	−0.27**	−0.02	-	
23. PHQ 9	−0.63**	0.72**	−0.02	−0.35**	−0.12†	−0.01	0.09	−0.58**	0.35**	−0.14†	−0.80**	-

specifying the processes involved in the purpose in life system, our model permits scientific testing, experimentation, and intervention. As such, our theoretical framework serves as the foundation for the development of the MMOP, a multidimensional measure designed to tap into the complex nature of purpose in life.

To evaluate the MMOP, we conducted two studies that examined its factor structure, reliability, and validity. Factor analyses from both studies indicated that the expected three-factor higher model (P-PIL, F-PIL, PCAT) showed the best fit to the data, outperforming both the single-factor model and the three-factor uncorrelated model. The MMOP items loaded well onto their respective subscale (>0.40), with large effects supporting our expectations that purpose in life is best captured by a multidimensional measure and that the MMOP is tapping into its distinct temporal features.

In Study 2, the MMOP demonstrated promising validity, showing significant positive correlations with other measures of purpose in life (LET and PIL-SF) and measures targeting related well-being constructs (curiosity, goal adjustment, positive affect, awareness of narrative identity, and hope). In contrast, and as predicted, the total MMOP correlated significantly and negatively with theoretically divergent constructs including goal disengagement, negative affect, anxiety, and depression further supporting the validity of the MMOP. Examination of divergent relationship of specific MMOP subscales showed that some negative relationships approached but did not reach statistical significance, namely between future oriented purpose in life and goal disengagement and between present oriented purpose in life and anxiety. The lack of statistical significance between the respective constructs may indicate that sample size limited power to detect smaller effects. Alternatively, it may reflect meaningful distinctions about how purpose in life functions. For instance, future oriented purpose in life may involve motivational processes that are unrelated to emotional disengagement from obstructed goals. In contrast, present purpose in life may reflect processes that do not correspond to anxious symptoms.

Despite some promising psychometric support for the MMOP, several limitations should be noted. First, the Study 2 factor analyses showed that the three-factor higher-order model fell short of meeting TLI and RMSEA model fit indices. This may be due to conceptual overlap between the MMOP dimensions, particularly between items measuring purpose congruence across time and both past and future oriented purpose in life. PCAT involves linking past, present, and future processes, captured in MMOP with questions such as '*Thinking about who I am today helps me to figure out how I want to live in the future'* (PCAT). We found this aspect is also reflected in some of the items measuring present oriented purpose (e.g., *I dedicate myself to things that feel meaningful*) and future oriented purpose (e.g., *I do things that are consistent with how I want to live my life*) given they aim to capture present and future oriented nature of purpose in life respectively. The high correlations (ranging from 0.75 to 0.93) between the ANIQ temporal dimension and the MMOP subscales further challenged the necessity of conceptualising purpose congruence across time as a dimension distinct from present oriented and future oriented purpose in life.

Although acceptable (Hair et al., 2009), the P-PIL subscale showed relatively modest internal consistency (0.68), possibly due to ambiguous item wording that led to varied interpretations by participants. For example, for Item 8 ("*discovering new things feels satisfying*"), some participants may have focused on their tendency to feel satisfied with new discoveries, while others might have thought about past discoveries or anticipated satisfaction from future discoveries. Additionally, reliance on a large university student population

in Study 1 (74%) and a predominantly Australian-based, Caucasian sample in both Study 1 and 2 (78% in Study 1 and 66% in Study 2) may limit the generalisability of findings. Therefore, the present findings should not be assumed to generalise beyond the Australian sample examined in this study.

Future research aiming to improve the MMOP should use clearer phrasing in the instructions and items to better highlight the temporal difference between items across the three dimensions. One possible approach is to begin each item with a phrase that appropriately orients the respondent toward the relevant timeframe. For instance, P-PIL items could start with the phrase …*when I reflect on the present*… to cue participants toward the present-oriented nature of P-PIL. While F-PIL items could start with …w*hen I reflect on the future*... to prompt a future-focused orientation. Furthermore, reconsideration of the MMOP's dimensions to reduce the redundancy of the PCAT dimension and place greater emphasis on past-oriented processes may contribute to a more balanced and comprehensive reflection of the purpose in life system. Incorporating qualitative feedback from participants could clarify interpretations and help identify ambiguities guiding further refinements to items. Future research should also examine the MMOP across culturally diverse populations with potentially different understandings and sources of purpose in life.

In conclusion, this paper presented a theoretically informed and holistic theoretical framework of purpose in life, viewing it as a dynamic system that integrates past, present, and future-oriented processes. Based on the purpose in life system, we developed the MMOP, which offers a temporally nuanced, multidimensional assessment of purpose in life. Unlike previous measures of purpose in life, the MMOP captures the extent people experience purpose in the present, future, and across time, enabling researchers and practitioners to identify specific areas of strength and/or challenges within individuals' purpose in life system. This detailed, process-oriented perspective allows for more precise examination of purpose in life, than what could be provided by a global purpose in life measure. Overall, the MMOP demonstrates promising factor structure, reliability, and validity. However, further refinement of the MMOP is needed to ensure stronger alignment with our theoretical framework and before the MMOP can be confidently applied in existential research and clinical practice.

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Data Availability The datasets generated by the research during and/or analysed during the current study are available in the Open Science Framework repository at https://osf.io/cswru/.

Declarations

Compliance with Ethical Standards Ethics approval for this project was granted by the Edith Cowan University Human Research Ethics Committee (REMS reference number: 2022-03883-FANG).

Informed consent All participants provided informed consent prior to their participation in this research.

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